

Resimulation Trigger Module (SRGM)

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Lifecycle & Resimulation Standard (SLRS)

Module: SRGM — Resimulation Trigger Module

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Governed Space: Resimulation Trigger Governance

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Canonical Definition

Resimulation Trigger Module (SRGM) is the third Core Module of the Simulation Lifecycle & Resimulation Standard (SLRS) within SIMULOS® — Simulation Governance Architecture.

SRGM governs the determination of when recognized Simulation Drift, material change, new evidence, changed simulation conditions, changed reality relationships, or other simulation-relevant lifecycle developments require a simulation to be executed again, updated before execution, partially repeated, comprehensively repeated, reconsidered, or escalated toward replacement or retirement assessment.

The Module transforms recognized lifecycle divergence into an explicit Resimulation Trigger State.

SRGM does not determine the detailed scope of the required resimulation. That responsibility belongs to SRSM — Resimulation Scope Module.

Its task is narrower and foundational:

to determine whether the existing governed simulation basis remains sufficient for continued simulation-specific use or whether a new simulation cycle must be initiated.

Architectural Position

SRGM is the third module within SLRS.

It follows:

SDRM — Simulation Drift Recognition Module

and precedes:

SRSM — Resimulation Scope Module.

The sequence is:

Lifecycle Monitoring → Drift Recognition → Resimulation Trigger → Resimulation Scope.

SLMM identifies materially relevant lifecycle changes.

SDRM determines whether those changes constitute Simulation Drift.

SRGM determines whether the resulting condition requires a new simulation cycle.

SRSM then determines what must actually be resimulated.

Internal Flow Position

Governed Simulation Drift Record Material Change Evidence Current Simulation Use Context → Trigger Candidate Identification → Trigger Relevance Assessment → Materiality Assessment → Existing Simulation Sufficiency Assessment → Continued-Use Consequence Assessment → Trigger Threshold Evaluation → Resimulation Trigger State → Governed Resimulation Trigger Record → SRSM — Resimulation Scope Module

Operational Space

SRGM governs the space between:

recognized simulation-relevant change or drift

and

the formal initiation of a new simulation cycle.

This operational space includes: trigger candidate identification; trigger source classification; drift-trigger relationships; material-change triggers; threshold-based triggers; event-based triggers; cumulative triggers; temporal triggers; evidence-based triggers; uncertainty-driven triggers; scenario emergence triggers; object-change triggers; model-change triggers; input-change triggers; correlation-change triggers; reliance-change triggers; trigger materiality; trigger urgency; trigger state; trigger escalation; trigger deferral conditions; trigger documentation.

Operational Scope

SRGM applies wherever previously governed simulations continue to be referenced, reused, compared, interpreted, relied upon, or incorporated into subsequent simulation, evidential, validation, engineering, scientific, mission, safety, or operational processes.

It may govern resimulation triggers for: individual simulation runs; repeated simulation programs; scenario portfolios; simulation campaigns; digital twins; engineering simulations; scientific simulations; mission simulations; safety simulations; infrastructure simulations; economic simulations; AI simulations; autonomous-system simulations; continuously updated simulations; long-lived simulation programs; future simulation technologies.

The Module remains technology-neutral.

Its concern is not how expensive or computationally difficult resimulation may be.

Its concern is whether continued use of the existing simulation basis remains methodologically supportable without a new simulation cycle. Drift Recognition
≠ Automatic Resimulation

SRGM establishes the fundamental distinction:

Recognized Simulation Drift ≠ Automatic Resimulation

Not every recognized divergence requires another simulation.

A drift condition may be: immaterial to the current use; outside the applicable scope; confined to an unused configuration; limited to an irrelevant scenario family; already bounded by an existing limitation; insufficiently developed to justify immediate resimulation; appropriate for continued monitoring.

Conversely, relatively small drift may require immediate resimulation where it affects a highly sensitive or consequential simulation condition.

The trigger must therefore be determined from the relationship between the change and the continuing use of the simulation.

Trigger Candidate Principle

Any lifecycle development capable of materially affecting the legitimate continuing use of a simulation may become a Resimulation Trigger Candidate .

Trigger Candidates may originate from: Simulation Object change; configuration change; reality change; representation change; assumption change; model change; simulation environment change; input or data change; scenario change; execution anomaly; output anomaly; correlation change; evidence degradation; changed reliance context; temporal expiration; newly discovered uncertainty; newly discovered failure mode; new operational observations; cumulative minor changes; compound Simulation Drift.

A Trigger Candidate is not yet a Resimulation Trigger.

It must first be assessed.

Trigger Source Classification

SRGM requires the source of a potential trigger to remain explicit.

A trigger may be classified as:

Object Trigger A material change to the Simulation Object.

Reality Trigger A material change in the reality represented by the simulation.

Representation Trigger A material deficiency or change in how reality is represented.

Assumption Trigger A material assumption has changed, weakened, failed, or become uncertain.

Model Trigger The simulation model has materially changed or become insufficient for continued use.

Environment Trigger The simulation environment or relevant simulation dependencies have materially changed.

Input Trigger The input basis or relevant data distribution has materially changed.

Scenario Trigger A new materially relevant scenario, edge case, stress condition, or scenario class has emerged.

Execution Trigger Execution behavior reveals anomalies, instability, drift, or conditions requiring renewed simulation.

Output Trigger Outputs reveal unexpected or materially unresolved behavior.

Correlation Trigger Simulation-to-reality correlation materially changes or deteriorates.

Evidence Trigger Simulation-derived evidence loses qualification, weight, applicability, or relevance.

Temporal Trigger A defined review period or simulation applicability period has expired.

Multiple trigger sources may coexist.

Material Change Trigger

A material change may require resimulation where it alters one or more conditions upon which the existing simulation results materially depend.

Examples include: new hardware configuration; software revision; changed physical component; changed environmental range; altered operating conditions; changed mission profile; changed behavioral logic; changed sensor characteristics; changed parameter range; changed infrastructure dependency.

The relevant question is not:

Did the object change?

It is:

Does the change materially affect the simulation basis upon which continued use depends?

Scenario Emergence Trigger

A new real-world or anticipated scenario may create a Resimulation Trigger even where the Simulation Object itself has not changed.

Examples include: newly observed failure combinations; new environmental extremes; new interaction patterns; new adversarial conditions; newly relevant edge cases; expanded operational domains; new mission phases; new human-system interactions.

Where the new scenario lies outside previously governed scenario coverage and materially affects the intended use of the simulation evidence, resimulation may be required.

Correlation Trigger

A previously acceptable simulation may require resimulation where new reality observations materially alter the established simulation-to-reality relationship.

Examples include: systematic real-world deviation; unexpected operational behavior; repeated prediction mismatch; degraded physical correspondence; changed environmental relationship; new empirical measurements.

SRCS governs the correlation relationship.

SRGM determines whether a material change in that relationship activates a new simulation cycle.

Evidence Degradation Trigger

Simulation evidence may become weaker even where the original simulation remains technically reproducible.

SERBS or lifecycle monitoring may identify: degraded evidential weight; reduced applicability; expired temporal relevance; new contradictory evidence; changed reliance conditions; weakened correlation; invalidated assumptions.

SRGM may treat such developments as Trigger Candidates.

Reproducible Simulation ≠ Permanently Sufficient Evidence.

Cumulative Trigger Principle

A series of individually non-triggering changes may collectively require resimulation.

Therefore:

No Single Trigger ≠ No Resimulation Requirement

SRGM must permit cumulative trigger assessment.

For example:

Minor Object Revision Small Input Distribution Shift New Edge Cases Moderate Correlation Deterioration

may collectively cross a resimulation threshold even though none independently would do so.

Compound Trigger Principle

Trigger conditions may interact non-linearly.

A changed assumption may become material only after a model update.

A new scenario may become critical only after an operational-domain expansion.

A modest correlation decline may become significant after reliance expands.

SRGM therefore evaluates both:

individual Trigger Candidates

and

combined Trigger Conditions.

Trigger Threshold

Organizations may define explicit Resimulation Trigger Thresholds.

Thresholds may be: qualitative; quantitative; probabilistic; event-based; consequence-based; sensitivity-based; time-based; configuration-based; evidence-based; hybrid.

SIMULOS® does not prescribe one universal numerical threshold.

The threshold must instead remain: explicit; relevant to the simulation purpose; connected to materiality; sensitive to the consequence of continued use; reviewable; traceable to its governance basis.

Trigger Sensitivity

Different simulation contexts may legitimately require different trigger sensitivity.

A minor change may justify resimulation where: simulation evidence supports safety-critical decisions; the model is highly sensitive to the changed variable; the simulation governs narrow tolerance conditions; correlation is already uncertain; consequences of unsupported reliance are severe.

The same change may not justify immediate resimulation for: exploratory

analysis; historical comparison; broad conceptual modeling; an unaffected scenario family.

Therefore:

Trigger Sensitivity Must Follow Simulation Relevance, Not Change Magnitude Alone.

Trigger Urgency

SRGM distinguishes whether resimulation is:

Non-Urgent Resimulation is required but continued bounded use remains temporarily acceptable.

Scheduled Resimulation must occur within a defined lifecycle or review window.

Priority Resimulation should occur before specified further reliance or decision activity.

Immediate Existing simulation reliance should not continue for the affected use until resimulation or another governed response occurs.

Urgency does not determine scope.

SRSM determines the required resimulation scope.

Resimulation Trigger States

SRGM may assign a governed Resimulation Trigger State. 1. No Trigger

Available change or drift does not require a new simulation cycle for the defined use. 2. Monitor

A Trigger Candidate exists, but current evidence does not justify resimulation.

The condition remains under lifecycle monitoring. 3. Conditional Trigger

Resimulation becomes required if a defined condition, threshold, event, use expansion, or additional change occurs. 4. Triggered

A governed basis exists for initiating resimulation. 5. Priority Trigger

Resimulation is required before specified further reliance, use, or lifecycle progression. 6. Immediate Trigger

The affected simulation basis should not continue to support the relevant use without immediate resimulation or another governed lifecycle response. 7. Indeterminate

Available information is insufficient to determine whether resimulation is required.

An Indeterminate state must not silently become No Trigger.

Trigger Deferral

A valid trigger may sometimes be deferred.

Deferral may occur where: required data are temporarily unavailable; a replacement model is being prepared; the affected simulation use has been suspended; the relevant operational condition will not occur before resimulation; another controlled mitigation temporarily bounds reliance.

Deferral must identify: reason; affected simulation; affected use; temporary limitations; expiration condition; review point.

Deferred Trigger $\neq$ Cancelled Trigger.

Trigger Closure

A Trigger Candidate or Trigger State must eventually reach a governed disposition.

Possible dispositions include: no resimulation required; continue monitoring; conditional trigger retained; resimulation initiated; trigger superseded; simulation replaced; simulation retired.

Unresolved Trigger Candidates must not disappear merely because time passes.
Resimulation Trigger $\neq$ Resimulation Scope

SRGM determines whether a new simulation cycle is required.

It does not determine whether the response requires: one repeated run; one scenario; one parameter region; one subsystem; one configuration; a partial campaign; a complete campaign; model reconstruction; simulation replacement.

That determination belongs to:

SRS — Resimulation Scope Module.

Therefore:

Trigger $\neq$ Scope. Resimulation Trigger $\neq$ Validation Trigger

A Simulation Resimulation Trigger concerns the need to re-enter the simulation lifecycle.

A Validation Trigger concerns broader validation governance.

A change may trigger: resimulation only; revalidation only; both; neither.

VALIDOS® remains responsible for broader validation lifecycle governance.

Negative Results as Trigger Inputs

Failed or anomalous simulations may themselves generate Trigger Candidates.

Examples include: unexpected instability; unexplained divergence; inconsistent repetitions; previously unseen failure; unexplained sensitivity; execution-specific anomalies; output patterns inconsistent with prior campaigns.

A failed execution does not automatically require resimulation.

But failure must remain available to trigger governance rather than being discarded as computational noise without assessment.

Traceability of Trigger Decisions

Every material trigger determination should remain traceable through:

Change Signal → Simulation Drift Record → Trigger Candidate → Affected Simulation Basis → Current Use Context → Materiality → Threshold → Trigger State → Urgency → Disposition → Resimulation Scope Assessment

This enables organizations to reconstruct not merely that resimulation occurred, but why it became necessary.

SRGM Boundary with SLMM

SLMM identifies lifecycle change.

SRGM does not independently replace lifecycle monitoring.

It receives relevant lifecycle information as trigger inputs.

Monitoring Signal ≠ Resimulation Trigger.

SRGM Boundary with SDRM

SDRM determines whether lifecycle change constitutes Simulation Drift.

SRGM determines whether that recognized drift requires a new simulation cycle.

Therefore:

Simulation Drift ≠ Automatic Resimulation.

SRGM Boundary with SRSM

SRGM determines whether resimulation is required.

SRSM determines what portion of the simulation lifecycle, scenario space, configuration space, or simulation campaign must be repeated or reconstructed.

Therefore:

Trigger Decision ≠ Scope Decision.

SRGM Boundary with SERBS

SERBS may identify degraded evidence, changed applicability, reduced evidential weight, or changed reliance boundaries.

SRGM may consume these conditions as Trigger Candidates.

It does not independently redefine simulation evidence qualification, weight, or reliance.

SRGM Boundary with SRCS

SRCS governs simulation-to-reality correlation.

SRGM may consume material correlation change as a trigger input.

It does not determine correlation itself.

SRGM Boundary with VALIDOS®

SRGM does not determine whether broader revalidation is required.

Its decision is simulation-specific:

Must a new simulation cycle be initiated?

VALIDOS® governs whether a Validation Object requires revalidation and what validation lifecycle response is necessary.

SRGM Boundary with ORA™

ORA™ governs Operational Reality.

Changes in Operational Reality may create Trigger Candidates where they affect the simulation basis.

SRGM does not determine or govern Operational Reality itself.

SRGM Boundary with INTEGROS®

INTEGROS® governs integrity properties.

An integrity issue may become a Trigger Candidate where it materially affects simulation relevance, but SRGM does not replace integrity governance.

SRGM Boundary with ADIT®

ADIT® may audit whether trigger evidence, decisions, thresholds, records, and lifecycle actions are preserved and verifiable.

SRGM governs the simulation-specific trigger determination itself.

Minimum Implementation Framework

A minimum SRGM implementation should establish: 1. Trigger Candidate Intake

Receive recognized Simulation Drift, material changes, new evidence, anomalies, correlation changes, and other simulation-relevant lifecycle signals. 2. Affected Simulation Identification

Identify the simulation, campaign, configuration, scenario family, evidence basis, or other simulation asset potentially affected. 3. Current Use Context

Determine how the existing simulation is currently being used, interpreted, or relied upon. 4. Trigger Relevance Assessment

Determine whether the Trigger Candidate can materially affect that continuing use. 5. Materiality Assessment

Assess the significance of the change relative to the simulation basis and intended reliance. 6. Cumulative and Compound Assessment

Evaluate whether multiple changes collectively create a trigger condition. 7. Trigger Threshold Evaluation

Compare the condition against the applicable qualitative, quantitative, event-based, temporal, or hybrid trigger criteria. 8. Trigger State Assignment

Assign an explicit Resimulation Trigger State. 9. Urgency and Deferral Control

Determine urgency and document any controlled trigger deferral. 10. Trigger Record Creation

Create a traceable record linking the trigger determination to its originating lifecycle conditions and downstream scope assessment. Use Case 1 — Space Mission Simulation: Mission Configuration Changes After Qualification Campaign

A spacecraft mission completes an extensive simulation campaign covering guidance, navigation, propulsion, thermal behavior, landing sequences, communication loss, contingency procedures, and mission-critical failure conditions.

The simulation evidence supports a defined mission configuration.

Months later, before launch, several changes occur: a flight-software update modifies guidance behavior; spacecraft mass distribution changes; a component is replaced; new environmental observations alter part of the landing-condition model; mission planners modify one phase of the descent profile.

None of these changes automatically proves that the previous simulation campaign is unusable.

Equally, the organization cannot safely assume that millions of previously successful simulation executions still represent the modified mission configuration.

SRGM receives the recognized drift conditions and determines whether they cross a governed Resimulation Trigger.

The Module asks:

Which changes materially affect the basis upon which the existing mission simulations continue to be relied upon?

A guidance-software modification affecting descent control may create a Priority Trigger.

A component change outside the simulated behavioral pathway may create No Trigger.

Updated environmental observations may create a Conditional Trigger pending assessment of whether the new conditions fall outside previous scenario coverage.

The result is not automatic repetition of the entire mission campaign.

It is a governed determination that establishes whether resimulation is required before deciding what must be resimulated.

This prevents two costly failures:

continuing to rely on stale mission simulations

and

repeating enormous simulation campaigns after changes that do not materially affect their basis. Use Case 2 — Autonomous Vehicle or Robot: New Real-World Failure Pattern After Deployment

An autonomous vehicle or mobile robot has completed millions of simulated scenarios before deployment.

After limited real-world operation, a previously rare interaction pattern begins appearing:

a combination of unusual lighting, partial sensor obstruction, human movement, and environmental geometry causes behavior that was not adequately represented in the original scenario portfolio.

The individual real-world events are uncommon.

However, they reveal: a new scenario class; changed understanding of sensor behavior; possible representation limitations; weakened simulation-to-reality correlation in the affected conditions; reduced evidential applicability of the original simulation campaign.

SRGM does not automatically order a complete rerun of millions of simulations.

Instead, it evaluates whether the new evidence creates a Resimulation Trigger.

If the condition affects a safety-relevant operational domain and lies outside previously governed scenario coverage, SRGM may assign an Immediate Trigger or Priority Trigger for the affected simulation basis.

The decision then passes to SRSM, which may determine that only:

the affected sensor conditions + scenario family + environmental range + related autonomous behaviors

require targeted resimulation.

The governance advantage is substantial.

Without a trigger framework, an organization may either dismiss rare real-world failures because the system passed millions of simulations or repeatedly rerun huge campaigns without knowing which part actually requires renewed testing.

SRGM creates the decision layer between those extremes:

new real-world evidence does not automatically invalidate all previous simulation, but material evidence must be capable of forcing simulation back into the lifecycle when the existing basis is no longer sufficient.

Governance Outputs

SRGM may produce: Resimulation Trigger Candidate Record Trigger Source Classification Affected Simulation Basis Record Current Simulation Use Context Record Trigger Relevance Assessment Trigger Materiality Assessment Cumulative Trigger Assessment Compound Trigger Assessment Trigger Threshold Record Resimulation Trigger State Trigger Urgency Classification Trigger Deferral Record Trigger Disposition Record Governed Resimulation Trigger Record Resimulation Scope Assessment Input

These outputs provide the governed basis for:

SRSM — Resimulation Scope Module.

Enterprise Application

SRGM gives organizations a systematic method for deciding when simulation must be performed again.

This becomes increasingly important as simulation programs grow larger, more expensive, more continuous, and more deeply connected to operational decisions.

Without explicit trigger governance, organizations risk two opposite failures:

under-resimulation — continuing to rely upon simulation after material conditions have changed;

and

over-resimulation — repeatedly executing expensive simulation campaigns after changes that do not materially affect the relevant simulation basis.

SRGM enables organizations to establish trigger logic around: product revisions; software updates; hardware changes; environmental changes; newly discovered edge cases; changed data distributions; model updates; new operational observations; degraded simulation-to-reality correlation; evidence degradation; changed reliance contexts; accumulated lifecycle drift.

For very large simulation campaigns, this can materially reduce unnecessary computational effort while ensuring that genuinely consequential change forces the simulation lifecycle to reopen.

SRGM therefore transforms resimulation from an ad hoc organizational judgment into a traceable governance decision.

Foundational Principles

Simulation Drift does not automatically require resimulation.

Absence of one decisive change does not mean absence of a cumulative trigger.

A Resimulation Trigger must be connected to the continuing use of the simulation.

Trigger sensitivity should reflect simulation relevance and consequence, not change magnitude alone.

New scenarios can trigger resimulation even when the Simulation Object has not changed.

New reality evidence can trigger resimulation even when the simulation remains technically reproducible.

Deferred resimulation is not cancelled resimulation.

An indeterminate trigger must not silently become No Trigger.

Trigger determination and resimulation scope are separate governance decisions.

The purpose of trigger governance is neither to maximize nor minimize resimulation, but to ensure that resimulation occurs when the existing simulation basis can no longer legitimately support continued use.

Canonical Closing Statement

Resimulation Trigger Module establishes the lifecycle decision layer of the Simulation Lifecycle & Resimulation Standard by transforming recognized Simulation Drift, material change, emerging scenarios, new evidence, correlation change, uncertainty, anomalies, temporal conditions, and cumulative or compound lifecycle developments into explicit and traceable Resimulation Trigger States. Through governance of Trigger Candidates, trigger sources, materiality, thresholds, sensitivity, cumulative and compound conditions, urgency, deferral, disposition, and traceability, SRGM prevents simulations from being repeated merely because change occurred while also preventing materially stale simulation bases from remaining in use because no single change appeared sufficient to force reconsideration. By separating lifecycle monitoring from drift recognition, drift recognition from trigger determination, trigger determination from resimulation scope, and resimulation from broader validation governance, SRGM creates the controlled decision point through which a governed simulation lifecycle may legitimately reopen. A simulation should be performed again not simply because reality changed, but because the change materially affects the basis upon which continued simulation use depends.

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